

Experiment - 7

Customer Segmentation using K-Means Clustering

```
In [1]: # Step 1: Load Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

df = pd.read_csv("Mall_Customers.csv")

print(df.head())

# Step 2: Explore Dataset

print("Dataset Shape:")
print(df.shape)

print("\nDataset Information:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Values:")
print(df.duplicated().sum())

# Step 3: Data Preprocessing

df = df.drop_duplicates()
df = df.dropna()

print("\nDataset after preprocessing:")
print(df.shape)

# Step 4: Select Relevant Features

X = df[[
    "Age",
    "Annual Income (k$)",
    "Spending Score (1-100)"
]]

print("\nSelected Features:")
print(X.head())
```

```
# Step 5: Feature Scaling

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

print("\nScaled Data:")
print(X_scaled[:5])

# Step 6: Elbow Method

inertia = []

for k in range(2, 11):

    model = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    model.fit(X_scaled)

    inertia.append(model.inertia_)

plt.figure(figsize=(8, 5))

plt.plot(
    range(2, 11),
    inertia,
    marker="o"
)

plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("Inertia")
plt.grid(True)

plt.show()

# Step 7: Silhouette Score

for k in range(2, 11):

    model = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    labels = model.fit_predict(X_scaled)

    score = silhouette_score(
        X_scaled,
        labels
    )
```

```
print(
    "K =", k,
    "Silhouette Score =", round(score, 4)
)

# Step 8: Apply K-Means Clustering

k = 5

kmeans = KMeans(
    n_clusters=k,
    random_state=42,
    n_init=10
)

df["Cluster"] = kmeans.fit_predict(X_scaled)

print("\nClustered Data:")
print(df.head())

# Step 9: Calculate Cluster Centroids

centroids = scaler.inverse_transform(
    kmeans.cluster_centers_
)

centroids_df = pd.DataFrame(
    centroids,
    columns=[
        "Age",
        "Annual Income (k$)",
        "Spending Score (1-100)"
    ]
)

print("\nCluster Centroids:")
print(centroids_df)

# Step 10: Visualize Customer Segments

plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df,
    x="Annual Income (k$)",
    y="Spending Score (1-100)",
    hue="Cluster",
    s=100
)

plt.scatter(
    centroids_df["Annual Income (k$)"],
    centroids_df["Spending Score (1-100)"],
    marker="x",
    s=250,
    c="black",
    label="Centroids"
```

```
)

plt.title("Customer Segmentation using K-Means")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score")
plt.legend()

plt.show()

# Step 11: Analyze Customer Clusters

cluster_analysis = df.groupby("Cluster")[
    [
        "Age",
        "Annual Income (k$)",
        "Spending Score (1-100)"
    ]
].mean()

print("\nCluster Analysis:")
print(cluster_analysis.round(2))

# Step 12: Count Customers in Each Cluster

print("\nNumber of Customers in Each Cluster:")

print(
    df["Cluster"].value_counts().sort_index()
)

# Step 13: Save the Final Dataset

df.to_csv(
    "customer_segments.csv",
    index=False
)

print("\nCustomer segmentation completed successfully.")
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Dataset Shape:
(200, 5)

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 200 entries, 0 to 199

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	CustomerID	200 non-null	int64
1	Genre	200 non-null	object
2	Age	200 non-null	int64
3	Annual Income (k\$)	200 non-null	int64
4	Spending Score (1-100)	200 non-null	int64

dtypes: int64(4), object(1)

memory usage: 7.9+ KB

None

Statistical Summary:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

Missing Values:

CustomerID	0
Genre	0
Age	0
Annual Income (k\$)	0
Spending Score (1-100)	0

dtype: int64

Duplicate Values:

0

Dataset after preprocessing:

(200, 5)

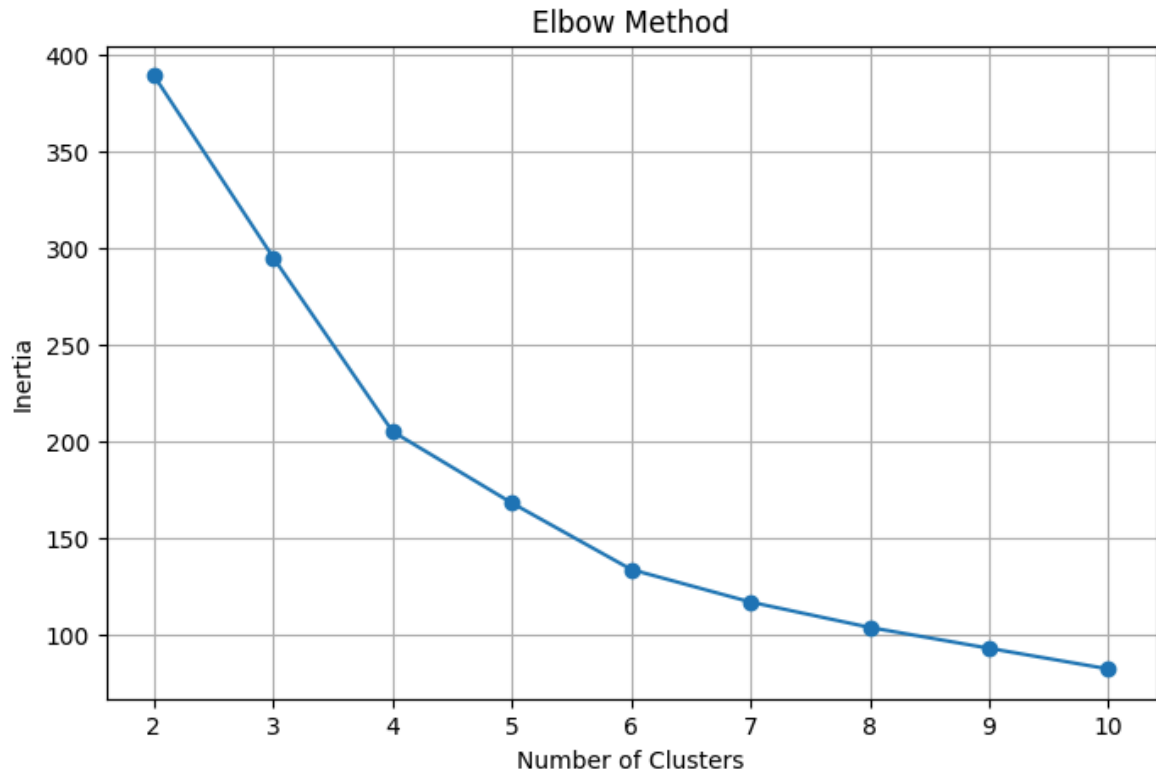
Selected Features:

	Age	Annual Income (k\$)	Spending Score (1-100)
0	19	15	39
1	21	15	81
2	20	16	6
3	23	16	77
4	31	17	40

Scaled Data:

```
[[-1.42456879 -1.73899919 -0.43480148]
 [-1.28103541 -1.73899919  1.19570407]]
```

```
[-1.3528021 -1.70082976 -1.71591298]
[-1.13750203 -1.70082976 1.04041783]
[-0.56336851 -1.66266033 -0.39597992]]
```



```
K = 2 Silhouette Score = 0.3355
K = 3 Silhouette Score = 0.3578
K = 4 Silhouette Score = 0.404
K = 5 Silhouette Score = 0.4166
K = 6 Silhouette Score = 0.4284
K = 7 Silhouette Score = 0.4172
K = 8 Silhouette Score = 0.4082
K = 9 Silhouette Score = 0.4177
K = 10 Silhouette Score = 0.4066
```

Clustered Data:

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	\
0	1	Male	19	15	39	
1	2	Male	21	15	81	
2	3	Female	20	16	6	
3	4	Female	23	16	77	
4	5	Female	31	17	40	

	Cluster
0	1
1	1
2	0
3	1
4	1

Cluster Centroids:

	Age	Annual Income (k\$)	Spending Score (1-100)
0	46.250000	26.750000	18.350000
1	25.185185	41.092593	62.240741
2	32.875000	86.100000	81.525000
3	39.871795	86.102564	19.358974
4	55.638298	54.382979	48.851064



Cluster Analysis:

	Age	Annual Income (k\$)	Spending Score (1-100)
Cluster			
0	46.25	26.75	18.35
1	25.19	41.09	62.24
2	32.88	86.10	81.53
3	39.87	86.10	19.36
4	55.64	54.38	48.85

Number of Customers in Each Cluster:

Cluster	
0	20
1	54
2	40
3	39
4	47

Name: count, dtype: int64

Customer segmentation completed successfully.